

# RSD-D

- Meets NEC 2017, 2020&2023 (690.12) requirements
- Executes rapid shutdown of system when Transmitter signal is absent
- Meets SunSpec requirements
- Dual-input channel

RSD-D meets SunSpec requirements, maintaining normal function by continually receiving a heartbeat signal from the APsmart Transmitter. The RSD-D executes rapid system shutdown when the Transmitter signal is absent. Users can manually execute rapid shutdown using Transmitter breaker switch.\*<sup>(1)(2)</sup>

## > RSD-D TECHNICAL DATA

### MODEL

### RSD-D-15

### RSD-D-20

#### INPUT DATA (DC)

|                                     |                   |                 |
|-------------------------------------|-------------------|-----------------|
| Range of Input Operating Voltage    | 8-65V Per Channel |                 |
| Maximum Cont. Input Current (Imax)  | 15A Per Channel   | 20A Per Channel |
| Maximum Short Circuit Current (Isc) | 25A               |                 |

#### OUTPUT DATA (DC)

|                                   |             |     |
|-----------------------------------|-------------|-----|
| Range of Output Operating Voltage | 16-130V     |     |
| Maximum Cont. Output Current      | 15A         | 20A |
| Maximum System Voltage            | 1000V/1500V |     |
| Maximum Series Fuse Rating        | 30A         |     |

#### MECHANICAL DATA

|                                         |                                                                                           |                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------|---------------------------|
| Operating Ambient Temperature Range     | -40 oF to +167 oF (-40 °C to + 75 °C)                                                     |                           |
| Dimensions (without cable & connectors) | 5.5" x 2" x 0.8"(140 mm x 50.6 mm x 20 mm)                                                |                           |
| Cable Length                            | Input 500mm/Output 2200mm                                                                 | Input 500mm/Output 2400mm |
| Cable Cross Section Size                | TUV:4mm²/UL:12AWG                                                                         |                           |
| Connector                               | Input: Stäubli MC4 PV-KBT4&KST4 or Customize<br>Output: APsystems specified or Customized |                           |
| Enclosure Rating                        | NEMA Type 6P/IP68                                                                         |                           |
| Protection Temperature                  | 100°C                                                                                     |                           |

#### FEATURES & COMPLIANCE

|                          |                                                                              |
|--------------------------|------------------------------------------------------------------------------|
| Communication Compliance | PLC                                                                          |
| Safety Compliance        | NEC 2017, 2020&2023 (690.12); UL1741;<br>CSA C22.2 No. 330-17; IEC/EN62109-1 |
| EMC Compliance           | FCC Part15; ICES-003                                                         |



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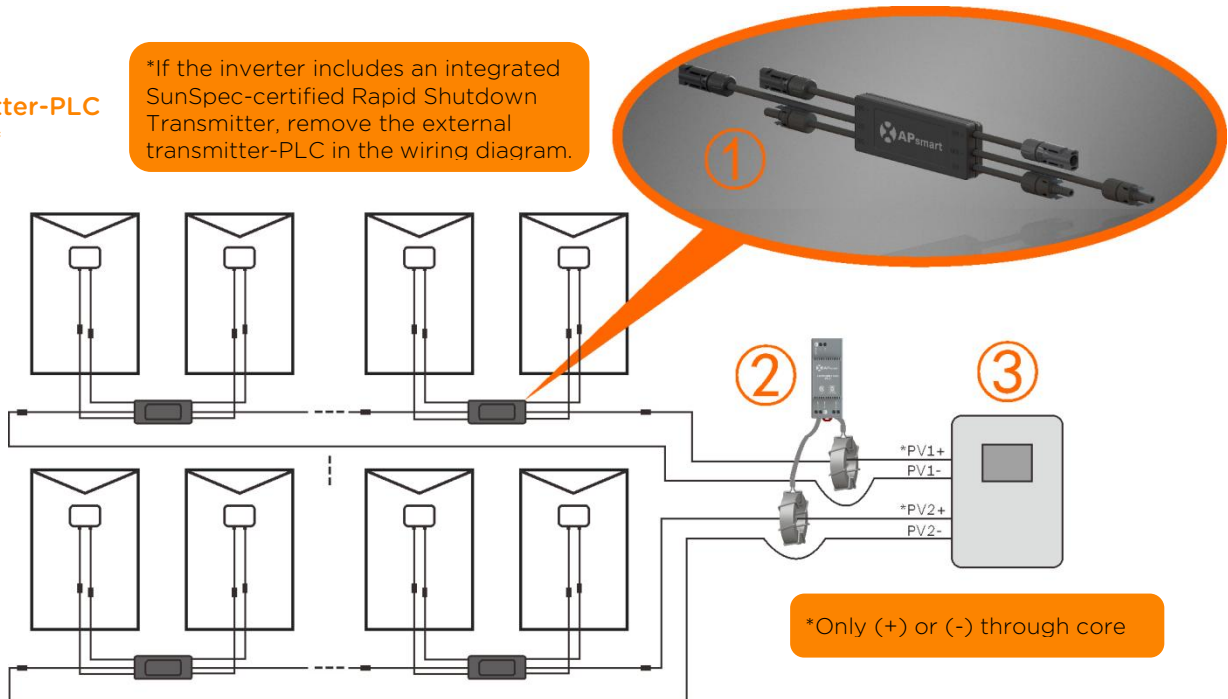
\*<sup>(1)</sup> RSD-D does not have automatic shutdown function for arc detection. When the system is abnormal, the transmitter signal is cut off by pulling the gate, which triggers shutdown.

\*<sup>(2)</sup> RSD-D is designed to reduce the risk of fire suppression but does not solve the risk of a arc fire.

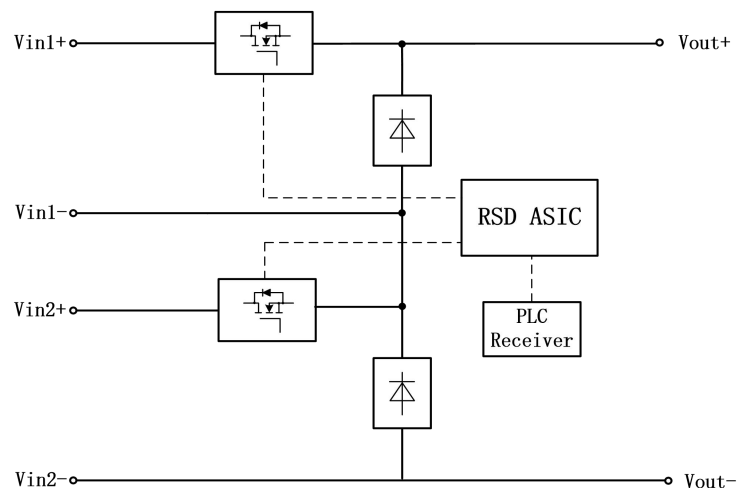
## > RSD-D WIRING DIAGRAM

- ① RSD-D
- ② Transmitter-PLC
- ③ Inverter\*

\*If the inverter includes an integrated SunSpec-certified Rapid Shutdown Transmitter, remove the external transmitter-PLC in the wiring diagram.



## > WORKING SCHEMATIC DIAGRAM



## ORDERING INFORMATION

|         |                                                               |
|---------|---------------------------------------------------------------|
| 426101  | 1500V UL/1000V TUV, 15A, 2.2m cable, Stäubli MC4 PV-KBT4&KST4 |
| 446101  | 1500V UL/1000V TUV, 20A, 2.4m cable, Stäubli MC4 PV-KBT4&KST4 |
| 4261xx* | 15A, 2.4m cable, Customize connector                          |
| 4461xx* | 20A, 2.4m cable, Customize connector                          |

\*please see the RSD Series Ordering Information